

# YASHRAJ KUMAR

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## Education

Indian Institute of Information Technology, Nagpur

B.Tech in Electronics and Communication Engineering

Jun. 2027

CGPA: 7.80 / 10

## Work Experience

YellowSense Technologies

Mar. 2026 – Apr. 2026

AI/ML Intern

Remote

- Architected a **7-layer liveness detection system** (FastAPI + Flutter) for contactless fingerprint capture using MediaPipe, OpenCV, HSV skin detection, and LBP texture analysis.
- Designed **anti-spoofing checks** to block fake fingerprints (silicone, latex, AI-generated) via reflection and ridge frequency analysis, achieving **100% spoof rejection rate**.

## Projects

🔗 **CodeSentinel AI – Developer Intelligence Platform** | Python, FastAPI, Groq, Qdrant, Redis, RAG, Docker May 2026

- Spearheaded an **AI code review platform** with a **6-stage pipeline**: static bug detection, LLM deep analysis, **RAG security scanning** against CWE database, auto-fix generation, and a **self-healing test loop** (3 retries).
- Pioneered a **codebase onboarding module** letting new developers ask natural language questions about any GitHub repo using **AST chunking**, call graph traversal, and **RAG on Qdrant**.
- Achieved **100% precision and recall** ( $F1 = 1.0$ ) on a 10-case security benchmark covering SQL injection, command injection, and hardcoded secrets with **zero false positives**.
- Deployed using **FastAPI** (4 endpoints), **Streamlit** UI, **Redis** caching, **Qdrant** vector search, and **Docker Compose** with 5 services.

🔗 **AI Mock Interview Coach** | Python, LangGraph, LangChain, Groq, ChromaDB, Streamlit, RAG

Mar. 2026

- Orchestrated an **AI mock interview system** with **LangGraph**, coordinating **4 specialized agents** (planner, question generator, evaluator, feedback) via a **stateful execution graph**.
- Integrated **RAG pipeline** with **conversational memory** for multi-turn context retention, enabling progressive difficulty scaling and coherent follow-up questions.
- Implemented **real-time evaluation engine** scoring candidates across **4 dimensions** (clarity, depth, relevance, overall) with **Pydantic-validated** structured output, generating reports in under **5 seconds**.
- Created **Streamlit interface** with adaptive prompt routing and **conditional graph edges** for agent-level orchestration across **6 interview categories**.

🔗 **YashGPT** | Python, Transformers, PEFT, LoRA, Mistral-7B, GPTQ, PyTorch, Hugging Face

Feb. 2026

- Fine-tuned **Mistral-7B-Instruct (GPTQ)** via **LoRA (PEFT)** on a custom dataset, training only **0.79% of parameters** (2.1M of 264M) as a virtual data science consultant.
- Reduced validation loss by **63%** ( $3.72 \rightarrow 1.39$ ) over 10 epochs using **FP16 precision**, **paged AdamW 8-bit** optimizer, and gradient accumulation.
- Optimized **7B parameter model** training via **gradient checkpointing** and **k-bit quantization**; deployed to **Hugging Face Hub**.

## Technical Skills

**Languages & Libraries:** Python, C/C++, SQL, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, OpenCV

**AI/ML & GenAI:** PyTorch, TensorFlow, LangChain, LangGraph, HuggingFace, Groq, Tavily, Ollama, RAG, LoRA/PEFT, Prompt Engineering, Agentic AI, Sentence Transformers, LangSmith

**Tools & Platforms:** Git/GitHub, Streamlit, FastAPI, Qdrant, ChromaDB, Redis, Vector Databases, Hugging Face Hub, Docker, Docker Compose, Workflow Automation, REST APIs, Webhooks

## Achievements & Certifications

- Secured **3rd position** at **Azeotropy ML Hackathon**, IIT Bombay, for building a model to predict infection rates of viruses like Ebola. 🏆
- Secured **Top 5** at **Predict2Protect ML Hackathon**, DMS IIT (ISM) Dhanbad, for a data-driven corporate bankruptcy prediction model.